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# What Makes a Patch?

The concept of the patch has mainly been used to map landscape ecologies in which humans have had little influence, such as forests and coral reefs; however, some authors have attempted to map Anthropocene geographies in relation to the ways humans have engineered varied spaces, such as cities, villages, and croplands.[1] This book updates these projects through attention to feral geographies: stimulated by human infrastructures but outside human control.

Patches can be identified at many scales, from a spot on a leaf to an ocean. The scale for any research project depends on the focus of the research; in choosing the correct scale, the researcher follows the phenomenon. The examples below should make this clearer. But before moving to examples, it seems important to say that, as an analytic tool, identifying patches cannot do everything. Identifying a patch does not reveal its social and ecological dynamics; that is a separate step. Still, it's a useful beginning in moving away from the homogeneous planetary space that possesses the popular imagination *as* the Anthropocene—and inspires many scholarly reactions, from the embrace of too-big-to-imagine "hyperobjects" to the condemnation of world-systems theorists.[2] A patchy Anthropocene is spatially heterogeneous, and this heterogeneity should be the object of our attention. Patches are a place to begin. Furthermore, to establish spatial heterogeneity, we can include ecological corridors—that is, lines along which organisms or nonliving things move—as a particularly linear form of the patch.[3]

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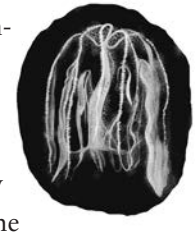


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Including corridors should not push us to de-emphasize places in favor of nonmaterial networks, such as those that connect experts and the objects of their expertise. Nor should we abandon places for discursive formations, that is, the clusters of participants, institutions, and ideologies that, as many social scientists have usefully shown, give power to places and the beings that happen to be there. Instead, we begin with the landscape ecology definition of the assemblage—as that collection of beings that happen to be in the patch—whatever their relations with each other and with humans. This kind of patch is usefully open-ended. It spurs our curiosity to ask, What's going on there?[4]

The advantages of designing research through the patch are most obvious when patch boundaries are rather definitive, as in a mainly contained body of water, such as the Black Sea. The Black Sea has outlets only at the Kerch and Bosporus Straits, on its north and southwest sides, respectively, as shown in Figure 11. The map shows currents within the Black Sea, which circulate water within the patch, with little interference from other bodies of water. These are the conditions under which the Black Sea became host to a great bloom of comb jellies in the late 1980s, as studied by the lab of Turkish marine biologist Temel Oğuz and reported by lab members Bettina Fach and Baris Salihoglu.[5] By 1989, they found, there was a "regime shift" in the organisms in the Black Sea. Instead of fish, there were only jellies. The jellies ate the fish when they were still tiny; the fish could not replace themselves.

What had happened? The containment of the sea is the key to their story. Once the Black Sea was a great fishery, but by the 1980s, the big fish had been removed through overfishing, and the fisheries concentrated only on anchovies. Meanwhile, runoff from commercial agriculture swept too many nutrients into the sea, causing eutrophication, which killed fish. With the amount of fishing as well as the deterioration of water quality, the fish were already in trouble. When comb jellies were accidentally introduced in the ballast water of incoming ships, and in the context of particular weather conditions, the jellies were able to take over—completely changing the system from dominance by fish life to dominance by jelly life. Fortuitously, a different jelly organism was accidentally introduced into the sea in the 1990s, and the new jelly began



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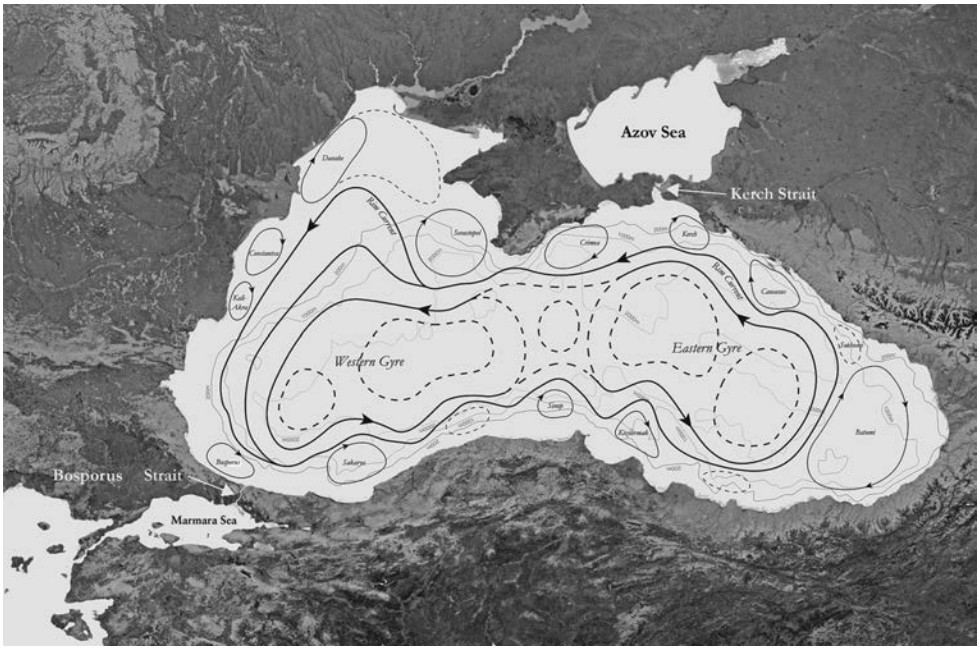



FIGURE 11. **The Black Sea is a deep, landlocked basin with exchange only to the Marmara Sea through the Bosporus Strait and to the Azov Sea via the Kerch Strait.**
LILI CARR

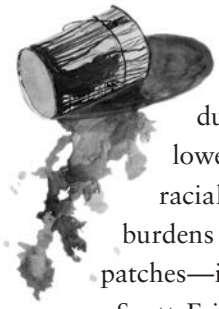
eating the original one. This allowed for at least some amount of biodiversity to return to the Black Sea.

The Black Sea has been, as the researchers put it, a "natural laboratory" for studying jelly invasion. The containment of the water meant that fish were not being replenished from the ocean, that excess nutrients were not being washed out to sea, and that when an exotic organism was introduced, the entire sea was its field of operation—allowing a complete takeover of the ecology. Obviously, there are connections that need to be considered, including the lack of regulation of interocean shipping, resulting in ballast waters full of potentially dangerous organisms, as well as the lack of regulation of agricultural runoff from surrounding lands. But in this contained space, the patch shows itself as a powerful tool for analysis.

Identifying the patch has been a key starting point in environmental justice research. Attention to the placement of industrial toxins, whether

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dumped or remaining from abandoned industrial sites, has allowed scholars and advocates to establish and confront unjust racial ecologies in which people of color are asked to bear the burdens of industrial poisoning. Identifying these toxic sites—as patches—is the first step.

Scott Frickel offers a study of brownfields—that is, abandoned industrial sites—many of which release toxins, especially in storms. In one instance, Frickel maps these sites in Rhode Island, and especially in its capital city Providence, where toxic substances seep and ooze from dozens of brownfields. As Frickel explains, "A fundamental feature of urban economies everywhere, industrial churning drives the incremental yet continuous spread of chemical hazards throughout urban areas . . . altering the chemical composition of city soils, water, and air."[6] Identifying brownfields is a descriptive task, necessary to understand cities. But there also is an ethical dimension. Frickel continues, "Recovering the lost history of these sites . . . gives city residents opportunities to reclaim a personal stake in their city's environmental history and its future." This is an issue for advocates as well as scholars. Social justice demands that we investigate unequal exposure to industrial toxins.

Environmental justice must be inside, not an add-on to, Anthropocene studies—a note that has recently been underscored by the call to address the uneven distribution of the impacts of climate change in terms of "critical climate justice."[7] Working with patches requires that we differentiate across ecologies—including those implicated in injustice. It is an important first step to understanding how landscapes are shaped into racial ecologies.

## Human Infrastructures Stimulate Patches of Feral Activity

To recap, in this book, infrastructures are projects that modify the land, the water, and the air. Infrastructures have feral effects, and these compose the Anthropocene.

In anthropology and science studies, infrastructures have most commonly been defined as networks that allow people, goods, and ideas to travel.[8] This allows an analysis of all the kinds of people that use a network, including those who don't like how the network functions.[9] Through network-based theoretical work, the literature on infrastruc-

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tures has usefully illuminated emergent political aspirations, dreams, and communities.

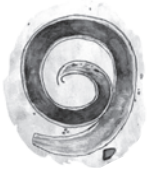
This framework is great for asking some questions, but not so great for others. We will return to political visions in Chapter 4, in asking why imperial and industrial infrastructures are built in repeating forms in so many places. Here, however, we need to see feral effects, which are obscured by this analytic framework. We need a more material definition.

Network analysis of infrastructure can attend to materiality—but only as it relates to the working of the network. This is because networks, which define infrastructure in such approaches, are not in themselves material; materiality must re-enter the analysis as it contributes to the network. Leakiness in the water system is inside the water-use network, but mosses living in the same water system (if not annoying to human water users) are not. The mosses may be part of a different network, but not the one defined by human water users. To take material arrangements outside the network seriously, it becomes necessary to return to the definition of infrastructure, to start with its material arrangements and forms, rather than its uses.

Feral effects are visible only when infrastructures are studied in their materiality—and as they make social and ecological patches and assemblages. As in network-oriented analyses, the material effects we chart are relational, but these relations are not confined by the designed function of the infrastructure and those who participate in it. For this work, then, infrastructures are building projects, not networks of users. In general, this is the definition used throughout this book.



The form and material features of infrastructure matter. Marine infrastructure, for example, provides underwater flat surfaces that are rare in nature. Jellyfish polyps, as studied by Martin Vodopivec and his lab, love these flat surfaces, forming great colonies through cloning, which mature into medusas that float across marine space. Marine sprawl increases the jellyfish population to boom—just because of its form.[10]



The feral effects of human infrastructure can be benign and ordinary. Zoologist Peter Funch writes about the meiofauna—tiny creatures—that live in urban drinking water, despite chlorination.[11] Bacteria are killed by the application of chlorine, but not these animals. Luckily, most people are not hurt by drinking and



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digesting them. But their presence is an example of nonhuman uses of infrastructure. Meiofauna would be at best peripheral participants in a network analysis of the water infrastructure (perhaps through the complaints of water users). In a material analysis beyond the users' network, they can take center stage. Their plentiful presence in human drinking water systems is an example of the infrastructure's feral effects—and a reason to focus on the material dynamics of infrastructure, as it makes patches for the livability of some beings but not others.[12]

The patch formation of meiofauna in urban water systems is a feral effect; the same array of species outside of the city might also form patches, but not patches of ferality. Thus, too, for invasive ants in California's cities: Ant biologist Deborah Gordon explains that invasive Argentine ants love people's houses, where there is just the kind of food and water they like.[13] But in wilder settings, native California ants fight back; the dominion of Argentine ants is far from guaranteed. Argentine ants hitchhiked on ships; they are encouraged by human housing. Their engagement with ships and houses creates their feral behavior; they are nonhuman users of human infrastructures. Thus, too, these ants turn out to be less of an ecological problem than imagined; they do not spread everywhere. Only in their relations to human infrastructures do these ants form Anthropocene patches.



Some feral effects are more dramatic, even in their patchiness. Staying with urban water, consider the arrival of rats in sewage systems, as described by historian Michael Vann.[14] Rats are not part of the intended purpose of sewage systems, but it turns out that rats flourish in the humid and nutrient-rich spaces of urban sewers.

Vann's research concerns the colonial construction of Hanoi in French Indochina. As in their other colonies, the French cleared space to build a beautiful modern city for white residents, while leaving the crowded remains to colonized Vietnamese residents. One of the hallmarks of the modern city was its European-style bathrooms, connected to an extensive sewage system. It was to be the pride of the colonial occupation, showcasing the wonders of civilization. Then the rats came. The modern sewers turned out to be an excellent habitat for rats, which thrived and multiplied. Crawling through the pipes, they emerged into the modern bathrooms of the French colonizers (Figure 12). The French were horrified, and they mounted campaigns in which



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FIGURE 12.
LIZ CLARKE.

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Hanoi residents were paid to bring rat tails to colonial authorities. Imagine their surprise when tailless rats began to crawl out of the sewers! The news that rats were even more prevalent in the white city than they were in the native city, because of the former's sewage system, was a blow to the pretensions of French colonialism. Like so many feral effects, rats in the sewers belie the utopian claims of modernization and development.[15] Vann's history opens the question of how European colonial expansion created a suite of feral effects, a question considered throughout the rest of this book. Here it seems useful to note that utopianism itself encourages feral effects by ignoring and overwriting the presence of local communities, human and nonhuman.[16]

## Feral Effects of the Plantation Form

One infrastructural project that tends to have dramatic feral effects is the monocrop commercial farm, the plantation. While the term "plantation" is the ordinary term for commercial tree-crop cultivation in the contemporary world, it also harkens back to the legacy of the European cultivation of exotic commercial crops around the world using coerced labor—including the New World's plantations using enslaved African people.[17]

This legacy of violence is useful to understand the tree-crop plantation today. The earlier plantation conditions helped create taken-for-granted features of the modern world, such as the discipline of labor to produce profits for distant investors.[18] The use of monocrops emerged in part from planters' counterfactual idea that the work could thus be done without much attention to the farming process. In fact, the enslaved people working the farms had considerable knowledge about plants, and this helped them survive under difficult circumstances.[19] Still, under plantation conditions, workers have often developed a hostile relation to plantation crops. Sidney Mintz, for example, explains that sugarcane workers in Puerto Rico in the 1950s described themselves as "going to battle" with the cane.[20] Because of the speed and discipline plantation owners demanded for cutting, the cane did, indeed, cause numerous worker injuries. Similar problems of alienation and injury are reported in contemporary tree plantations. Tania Li and Pujo Semedi found that

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oil palm plantations in West Kalimantan, Indonesia, were established with a kind of racialized discipline in which workers were sacrificed to efficient production.[21] Furthermore, contemporary commercial tree plantations continue to be associated with the violence of dispossession, as Indigenous residents are removed from their traditional territories.[22] The term "plantation" has returned, indeed, to describe contemporary land grabs, involving all kinds of crops.[23]

It is useful to reproduce Li and Semedi's respective definitions of the plantation—as exemplified in Indonesia—to get a sense of the term's contemporary traction.[24]

> Pujo: A plantation is a giant, an inefficient and lazy giant, but still a giant. It takes up a huge amount of space. It is greedy and careless, destroying everything around. It is alien, strange, and unpredictable. It is human, but you cannot form a normal human relationship with it. It can trample you, eat you, or drain your strength then spit you out. It guards its treasure. You cannot tame it or make it go away. You have to live with it. But it is a bit stupid, so if you are clever you can steal from it.
>
> Tania: A plantation is a machine that assembles land, labor, and capital in huge quantities to produce monocrops for a world market. It is intrinsically colonial, based on the assumption that the people on the spot are incapable of efficient production. It takes life under control: space, time, flora, fauna, water, chemicals, people. It is owned by a corporation and run by managers along bureaucratic lines.

Through its numerous legacies of violence, the plantation thus straddles and connects imperial and industrial legacies. In the process it creates dangerous Anthropocene patches. Chapter 1 discussed toxic chemicals on Philippine banana plantations, which threaten local people, their animals, and their crops. Monocrops also attract and nurture diseases, which can spread far beyond the plantation to destroy what otherwise might be much more sustainable peasant farms.

Agroecologist Ivette Perfecto has studied this problem in the coffee plantations of Central America.[25] It's possible to grow coffee under varied conditions, as it does quite well in the shade of other trees. Peas-

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ants in Central America have a history of growing coffee together with other tree crops in polyculture food orchards. However, in the last few decades, monoculture, full-sun coffee plantations have also made an appearance. Figure 13 shows the varied ways coffee can be grown in this area, which range from allowing coffee to be an element in an anthropogenic forest (as shown at the top of the figure) to full-sun monoculture (at the bottom).



It turns out that these varied methods of growing coffee have very different effects in relation to coffee diseases. Coffee rust fungus is a devastating coffee disease that is spread by wind-borne spores. Planting coffee in the shade of other trees offers considerable protection from coffee rust fungus. In contrast, full-sun monoculture allows spores to gather, launching an attack of the disease. Furthermore, once the disease is established in a particular place, it becomes a source for infection across the whole region, creating an ever-growing patch of infection. At some point, the disease becomes so saturated in the region that it can no longer be controlled; it infects peasant polyculture farms as much as monocrop plantations. At this point, it becomes an epidemic. No trees are safe. Furthermore, just planting more shade trees no longer solves the problem. This is a "state change": a transformation of the ecological dynamics of the region. Perfecto explains the problem through the concept of hysteresis. Once a state change has occurred, it becomes very difficult to go back to the original state. Coffee may no longer be a viable crop.[26]

As mentioned in the Introduction, this book singles out imperial and industrial infrastructures, rather than all human infrastructures, to explain the Anthropocene. We argue that the scale of imperial and industrial infrastructure building in the last five hundred years has caused more systems ruptures—that is, state changes—than ever before in human history. The concept of state change suggests how deeply these infrastructures can affect landscapes. The whole matrix of peasant food crops and agroforestry as well as their cash crops are affected; a new patch structure is coming into being. This poses problems for understanding time as well as space. The discussion of state changes will be continued in Chapter 4.

What are the best ways to illustrate the feral dynamics of Anthro-

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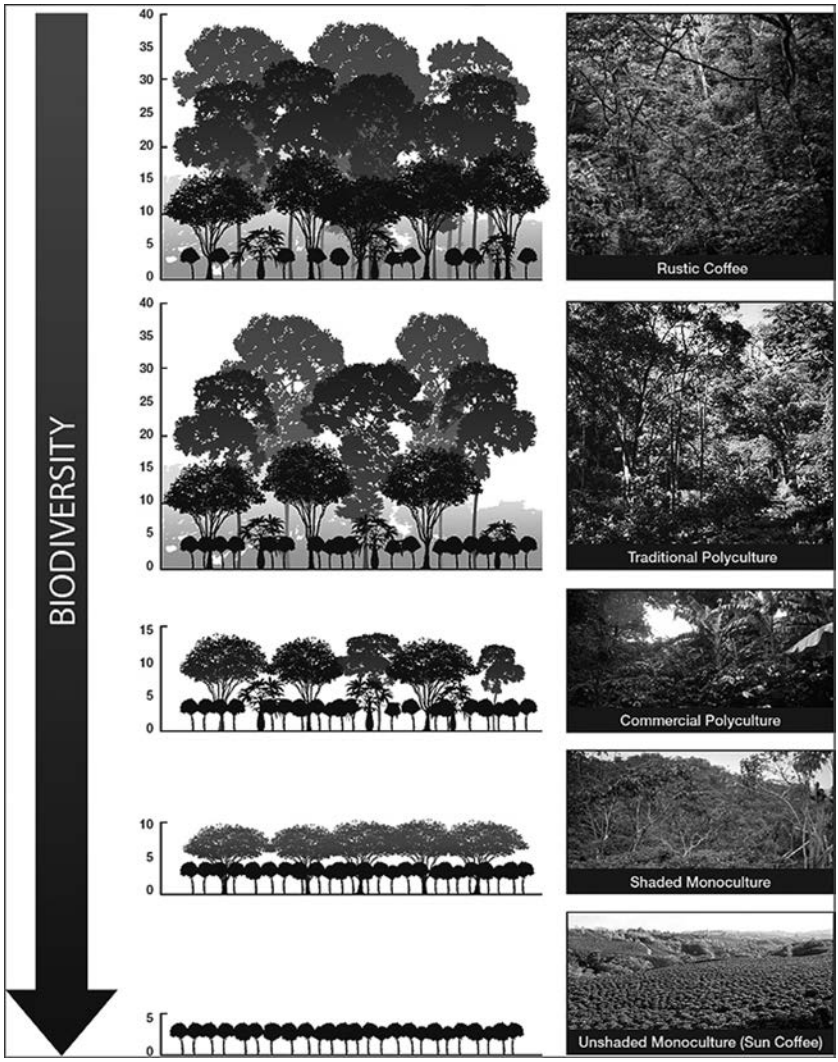



FIGURE 13. **Coffee intensification (technification) gradient.**
DAVE BRENNER

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pocene patches? This book has already offered a variety of images and diagrams as aids to explaining patch-based feral effects. Only a few are easily identifiable as "maps." Yet each aids our ability to analyze the patch as a site of feral action. We now turn to further discussion of maps as ways to investigate and describe Anthropocene patches.

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